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RESEARCH

$CD4^-/CD8^-$ double-negative tumor-infiltrating lymphocytes expanded from solid tumor tissue suppress the proliferation of tumor cells in an MHC-independent way

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Abstract

Purpose Tumor-infiltrating lymphocytes (TILs) have shown remarkable clinical responses in some patients with advanced solid tumors. As a rare subset of TILs, $CD4^-/CD8^-$ double-negative T cells (DNTs) were poorly known. This study aims to investigate the characteristics and function of $CD3^+CD4^-CD8^-$ TILs (double-negative TIL, DN-TILs) derived from solid tumor.

Methods DN-TILs were derived and expanded ex vivo from resected gastric carcinoma tissue and phenotyped by flow cytometry. The cytotoxicity of DN-TILs was determined against established tumor cell lines in vitro or through in vivo adoptive transfer into xenograft models. K562 cells were transferred with the HLA gene to verify whether the cytotoxicity of DN-TILs was MHC-independent.

Results Flow cytometric analysis revealed a high-purity population of DN-TILs (> 97%) within $CD3^+$ TILs, which expanded more than 800-folds in 2 weeks, consisting of a mixture of alpha–beta (αβ) and gamma–delta (γδ) T-cell receptor (TCR)—expressing cells (with the majority being αβ-TCR, > 95%). Using single-cell RNA sequencing, the expanded DN-TILs were categorized into four main subsets, Natural Killer T cells (approximately 80%, 5563 in 7028), Progenitor cells, Germ cells and T helper2 cells. DN-TILs exhibited a broad anticancer cytotoxicity in a donor-unrestricted manner against various cancer cell lines derived from pancreatic cancer (Panc-1), gastric cancer (HGC-27), ovarian cancer (SKOV-3), malignant melanoma (A375). The cytotoxicity was MHC-independent, which was not altered in K562 transferring with HLA gene or not. DN-TILs significantly reduced tumor volume in xenograft models with superior tumor-homing ability and low off-target toxicity.

Conclusion Gastric carcinoma derived DN-TIL can target tumor cells in vitro and in vivo. DN-TILs have the potential to be used as a adoptive cell therapy for solid cancers with both the advantages of DNT and TIL.

Keywords Double negative T cells · Tumor-infiltrating lymphocyte · MHC-independent · Adoptive cell therapy · Solid tumors

Introduction

Since the first reports describing treatment with in vitro expanded tumor-infiltrating lymphocytes (TILs) in 1986 (Rosenberg et al. 1986), TILs have shown impressive results in patients with advanced solid tumors (Sarnaik et al. 2021; Stevanović et al. 2019; Creelan et al. 2021). However, TIL therapy is a highly personalized treatment with unique properties for each individual patient (Berg, et al. 2020; Poschke et al. 2020). With the ever-increasing complexity in understanding TIL composition and localization, $CD3^+CD4^-CD8^-$ double-negative (DN) T cell as a rare subset of TILs was concerned (Paijens et al. 2021).

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DNT cells only count for 1–5% in peripheral blood T cells (Lee et al. 2018; Fischer et al. 2005; Merims et al. 2011) but play important roles in GvHD, autoimmunity disease and oncotherapy (Wu et al. 2022). DNT isolated from peripheral blood was proved to inhibit the proliferation and invasion of human pancreatic cancer cell PANC-1 through the expression of IFN-γ and FasL (Lu et al. 2019). Different from TCR of $CD4^+$ T and $CD8^+$ T, which is mainly αβ, the TCR of DNT contains both αβ and γδ, so it can participate in both innate immunity and acquired immunity. Moreover, the anti-tumor function of DNT both in vivo and in vitro is independent of the MHC limitation (Alnaggar et al. 2019; Chitadze et al. 2017; Kabelitz et al. 2007; Vantourout and Hayday 2013).

It has been reported that $TCR\alpha\beta^+$ DNT cells account for 1.4% of $CD3^+$ T cells within the tumor tissues of non-small-cell lung cancer (NSCLC) patients (Stankovic et al. 2018). It is difficult to expand the rare DN-TIL. Therefore, the features of DN-TIL are not clear.

In this study, we used flow cytometry to explore its features on the cell surface and single-cell sequencing to define the exact cell subsets. K562-HLA was constructed to verify its cytotoxicity independent with MHC. RTCA assay on HGC-27, PANC-1, SKOV-3 and A375 and tumor-bearing mice model were conducted to explore its cytotoxicity on tumor cells.

Materials and methods

Culture of TIL

Resected human gastric carcinoma tissue from patient were collected upon informed patient consent approved by the ethics committee of Shanghai Tenth People's Hospital, School of Medicine, Tongji University, Shanghai (Ethical code, SHSY-IEC-4.1/20-230/03). Fresh tumor tissues were transported to the laboratory by cold chain in tissue preservation solution containing 1% penicillin and streptomycin. After thoroughly cleaning with saline, removing adipose and necrotic tissue, the tissue blocks were cut into 1–3 mm^3, and 20 pieces were put per well in the G-Rex culture dish for growing with X.ViVo, 5% AB serum and 300 IU/mL interleukin (IL)-2 (201-GMP-01M, R&D).

Before counting, 10 μL cells were taken, mixed with 10 μL AO/PI dye, and placed in the chamber of the counting plate. Cell density was measured every 3 days using a Countstar automatic cell counter and multiplied by the cell volume to obtain the total number of cells. The current number of cells divided by the initial one results in a multiple of cell expansion.

DN-TIL was activated in a plate pre-coated with anti-CD3 (OKT3) (16-0037-81, ThermoFisher) and anti-CD28 Monoclonal antibody (CD28.2) (16-0289-81, ThermoFisher) in 5% CO_2, 37℃ for 2 h before further use.

Flow cytometry

DN-TIL cells were washed and resuspend in 200 μL PBS. FITC anti-human CD3 Antibody (317326, Biolegend), PE/Cyanine7 anti-human CD45 Antibody (304015, Biolegend), PerCP/Cyanine5.5 anti-human CD4 Antibody (300529, Biolegend), APC/Fire™ 750 anti-human CD8 Antibody (344745, Biolegend), Purified anti-GFP Antibody (338001, Biolegend) were added to DN-TIL cells and incubated for 30 min in dark place before detection. Flowjo Software was used for detection and analysis.

Single-cell transcriptomic sequencing

Seurat (version 4.3.0) R package (version 4.1.1) was used for single cell data analysis following the standard protocol without any unnecessary modifications (Stuart, et al. 2019). Data were normalized with NormalizeData function. Feature counts of each cell were divided by the total counts for that cell multiplied by a scaler factor (1E4), then natural-log transformed. The normalized data were then integrated for uniform manifold approximation and projection (UMAP) clustering. The mitochondrial DNA exclusion cutoff in the R diagram was set at −0.25. After quality control and filtering, we performed an integrated analysis and identified 4 clusters of cell subpopulations. FindMarkers (min.pct = 0.25) function was used to find cell markers for each cluster. Clusters were annotated based on canonical cell markers (http://xteam.xbio.top/CellMarker/).

Cells

The human pancreatic cancer cell line Panc-1, the human ovarian cancer cell line SKOV-3, the human malignant melanoma cell line A375, human chronic myeloid leukemia cells K562 were purchased from American Type Culture Collection (ATCC). The human gastric cancer cell line HGC-27 was purchased from GENECHEM Medical Technology Co., LTD. The cells were cultured in a humidified incubator with 5% CO2 at 37 °C. All the medium were supplemented with fetal bovine serum (FBS), 100 U/mL penicillin and 100 μg/mL streptomycin. Panc-1 and A375 were maintained in DMEM with 10% FBS. HGC-27, K562 and SKOV-3 were maintained in RPMI1640 (20% FBS), IMDM (10% FBS), and McCoy's 5A (10% FBS) respectively. The PBMC was purchased from Shanghai Sailybio Technology Co., Ltd., and HLA type is A0201, article number is SLB-HP050B, 5E7/piece.

Cytotoxicity assay

HGC-27, SKOV-3, PANC-1 and A375 were put in the RTCA E-Plate Insert S16 plate (xCElligence RTCA S16, ACEA Biosciences) 24 h before effector cells. The cell index will increase when the tumor cells grow and stick to the bottom of the E-plate, and decrease when they are killed and leave the bottom. When the cell index raised to around 1.0, the effector cells were put into the selected wells in different Effector: Target (E:T) ratio. The time and cell index graph of each plate was draw automatically by the xCELLigence realtime cell analyser (RTCA) system (ACEA Biosciences, San Diego, California, USA) to react the growth of target cells which indicated cytotoxicity of effector cells indirectly. Each experiment was repeated three times, with three wells for each condition.

Transferred HLA to K562 cells

Lentivirus system was used to generate stable expression of K562-EGFP cells and K562-EGFP-HLA-A0201 cells. The HLA over or none expression vector carrying the GFP reporter gene was synthesized by Gikai Gene Biology, and the virus titer was determined after large-scale amplification, concentration, and purification through HEK293T cells (ATCC, catalog: CRL-3216). K562 cells were transfected with EGFP or EGFP-HLA-A0201 lentivirus with MOI = 10 and stably expressed cell lines were selected using puromycin.

The successful transfection of K562 cells was determined by flow cytometry. 4×10^5 K562-EGFP and K562-EGFP-HLA-A0201 cells were cocultured with DN-TIL and PBMC cells, respectively, in an effector: target ratio of 1:2, 1:1, 2:1, 5:1, 10:1 for 48 h. The microscopic and electron microscopic photographs were taken at 0 and 48 h of coculture. The ratio of living K562 cells and T cells was determined by flow cytometry.

Xenograft model

6–8 weeks female NOD. CB17-Prkdcscid Il2rgtm1/Bcgen (B-NDG) mice were purchased and housed in BioDurosundia Vivarium (Shanghai) Individual Ventilation Cages at constant temperature and humidity with Maximum 6 animals in each cage. The situation they live in is pathology-free. All in vivo experiments were performed in accordance with the established institutional guidelines. The Animal Care Committee of Tongji University School of Medicine approved the study protocol. Animals were not allowed to become moribund.

1E7 EGFP pre-labeled HGC-27 cells in 200 μL RPMI-1640 were given subcutaneously to induce tumor growth. When tumor volume reached at about 53 mm^3, 12 mice were further divided into 2 groups (six mice per group) randomly and treated with PBS and 2E7 DN-TIL in 200 μL X.ViVo, respectively in D0, D3 and D6. Body weight was recorded every 2 days. In vivo imaging system (IVIS) imaging was taken once a week to evaluate the killing effect. To evaluate tumor load, tumor weight was recorded and tumor volume was measured using vernier calipers and calculated using formula $A \times B^2 \times 0.5$, where A and B represented the longest and shortest diameters of the tumor, respectively. After 3 weeks, mice were sacrificed and solid tumors were obtained for pathology and IHC. Major organs such as liver and spleen were examined for pathology and DN-TIL infiltration (Ramachandran et al. 2019).

Immunohistochemistry

Tumor tissue from mouses was fixed in 10% formalin and embedded in paraffin. Dewaxing from xylene to 70% ethanol, repair the antigen in a boiled citrate solution following the manufacturer's instructions. Immunohistochemistry was performed using primary antibody CD3ε (D7A6E™) XP® Rabbit mAb (Cell Signaling Technology, #85061).

Statistical analyses

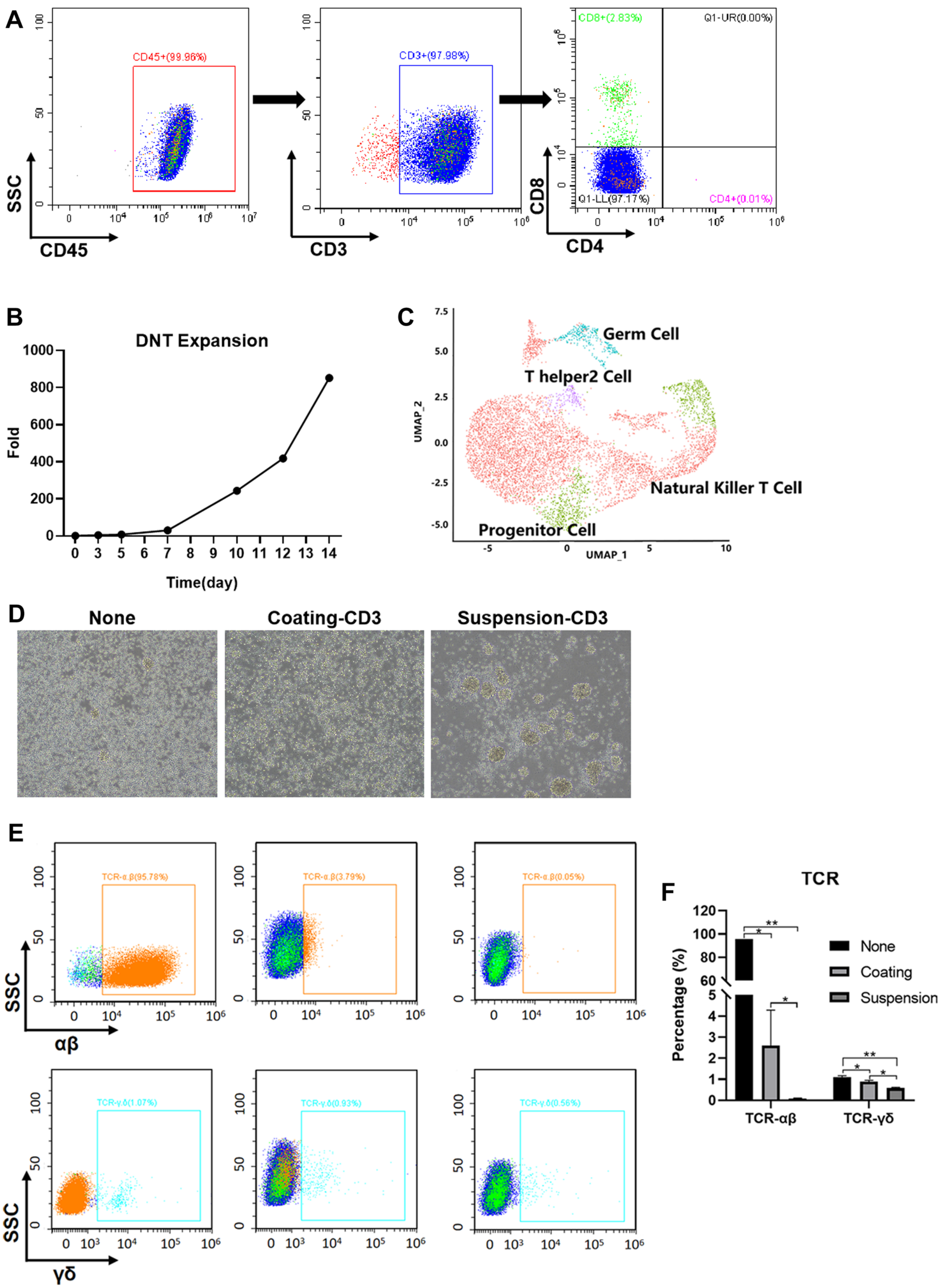
Data analysis was performed using GraphPad Prism 8.0.2. To assess the normality of the data, the Shapiro–Wilk test was employed with a significance level of $P > 0.05$. The results were presented as mean ± standard deviation (SD). Student T test was used to compare two groups and a P value < 0.05 was considered statistically significant. The notation "*", "**", "***", and "****" was used to represent P values of < 0.05, < 0.01, < 0.001, and < 0.0001 respectively.

Results

Characteristics of DN-TIL

Figure 1a shows a high DN-TIL purity (97.17%) in $CD3^+$ TILs. DN-TIL expanded almost 800-folds in about 2 weeks with a sharp upward trend following the same culture scheme of TILs (Fig. 1b).

Single-cell transcriptomic sequencing was conducted to define the cell subpopulations by cell markers. And DN-TILs were divided into four major clusters as Natural Killer T (NKT) cells accounting for the majority of all cells (around 80%, 5563 in 7028), Progenitor cell, Germ cell and T helper2 cell (Fig. 1c). Table 1 shows the cell markers in detail. Compared to suspended activation, DN-TILs in precoating activation with anti-CD3 and anti-CD8 antibodies had higher viability and a more uniform state instead of


A
CD45+(99.96%)
SSC
CD45
CD3+(97.98%)
CD3
CD8+(2.83%)
Q1-UR(0.00%)
Q1-LL(97.17%)
CD4+(0.01%)
CD8
CD4
B
DNT Expansion
Fold
Time(day)
C
Germ Cell
T helper2 Cell
Natural Killer T Cell
Progenitor Cell
UMAP_2
UMAP_1
D
None
Coating-CD3
Suspension-CD3
E
SSC
αβ
γδ
F
TCR
Percentage (%)
None
Coating
Suspension
TCR-αβ
TCR-γδ

◂**Fig. 1** Characterization of double-Negative Tumor-infiltrating lymphocytes (DN-TIL). **A** Derived garstric DN-TILs were characterized phenotypically by flow cytometry for CD4 and CD8 expression on $CD3^+$ lymphocytes; **B** Expansion profile of DN-TILs expanded under GMP conditions over 2 weeks; **C** UMAP plot of 7028 DN-TIL s that were segregated into four clusters; **D** DN-TILs were activated by anti-CD3 and anti-CD8 in different ways, pre-coating or suspending (×100 magnification); **E** Pre- and post-activated DN-TILs were phenotyped by flow cytometry for TCRαβ and TCRγδ expression; **F** statistical chart of differences. *$P<0.05$, **$P<0.01$

gathering together (Fig. 1d). Almost all the expanded DN-TILs expressed TCRαβ (> 95%) which decreased to 3.79% after activation (Fig. 1e).

DN-TILs suppressed the proliferation of multiple tumor cell lines in dose-dependent manner

RTCA assay was used to determine the cytotoxity of DN-TILs. Representative images showed that DN-TILs could inhibit the proliferation of gastric cancer line (HGC-27) and pancreatic cancer cell line (PANC-1) in the digestive system, malignant melanoma cell A375 in soft tissue sarcoma, and ovarian cancer SKOV-3 in gynecological tumors. And inhibition of DN-TIL was positively correlated with the effector: target ratio (Fig. 2) in SKOV-3, A375, and HGC-27 ($P<0.05$), while there was no statistical significance in PANC-1.

DN-TILs suppressed the proliferation of K562 in an MHC-independent way

To verify whether DN-TILs have the same MHC-independent killing ability as DNTs, we transferred HLA to K562 cells that did not express MHC. After 48-h incubation, we found that there was no significant difference in the killing effects of DN-TILs cells on K562 and K562-HLA cells, while the killing effects of PBMC on K562-HLA were significantly increased than in the K562 group (Fig. 3).

DN-TILs suppressed the proliferation of xenograft on tumor-bearing mice

We also observed the inhibitory effect of DN-TILs on HGC-27 derived xenografts in tumor-bearing mice. From day 20 of tumor transplantation, total flux changes (Fig. 4a, b) and tumor weight (Fig. 4e) showed no significant difference between the two groups with a lower trend in DN-TILs group. The tumor volume showed significant decreases between DN-TILs group and PBS group on day20 (897.26 ± 83.98 mm^3 vs. 1366.13 ± 108.09 mm^3, $P<0.001$), day24 (1338.93 ± 103.99 mm^3 vs. 1950.34 ± 162.71 mm^3, $P<0.001$) and day25 (1391.42 ± 106.22 mm^3 vs. 1989.68 ± 400.80 mm^3, $P<0.001$) (Fig. 4c). Interestingly, IHC showed that DN-TILs infiltrate in tumor tissues only (Fig. 4f).

Discussion

TIL is composed of T cells with multiple T cell receptor (TCR) clones capable of recognizing an array of tumor antigens (Schumacher and Schreiber 2015; Titov 2021). Therefore, compared to other adoptive cell therapy (ACT), TIL therapy has its unique advantages. And the DNT cell was a rare unconventional subset of T cell. DNT cells have been shown to infiltrate solid tumors such as NSCLC, liver cancer, glioma, and pancreatic tumors (Fang et al. 2019; Di Blasi et al. 2020; Liu et al. 2017; Hall et al. 2016). To our knowledge, this is the first study on the DN-TILs derived from gastric carcinoma.

DN-TIL in solid tumors remains largely unexplored, possibly due to the difficulty of expansion induced by low frequency (Fang et al. 2019). Interestingly, in our study, a high percentage and purity of DN-TILs were obtained (> 95%, Fig. 1a). And this group of DN-TILs was expanded to almost 800 times in a 2-week conventional culture regimen (Fig. 1b).

TIL subsets are relatively conservative, even among different tumors in different patients. On the contrary, the relative proportions of each subgroup of TILs showed considerable heterogeneity. The previous study found that macrophages accounted for the highest proportion of intrusions into colorectal cancer, followed by $CD4^+$ T cells, mDC cells, $CD8^+$ T cells, neutrophils, B cells, monocytes, and NK cells (Qin et al. 2022). While our analysis of DN-TILs identified four major clusters as NKT (80%), Progenitor cell, Germ cell and Th2 cell (Fig. 1c). There is great heterogeneity in the tumor microenvironment among different tumors, which may lead to alteration of infiltrating level of TIL subgroups. The different subset and proportion potentially suggest that the balance between different tumors with immune cell states is a major factor in regulating the anti-tumor immune response.

In both humans and mice, approximately 95% of peripheral blood T cells express TCRαβ, whereas 5% of T cells express TCRγδ (Creelan et al. 2021). While DNT cells contain both TCRαβ T cells and TCRγδ T cells in different proportions according to various studies. A large proportion of expanded DNTs are γδ T cells basing on their relatively high abundance in the peripheral blood (Fisher et al. 2014). As seen in Yao's study, the majority of expanded DNTs contained a mixture of αβ-(~ 10%) and γδ-T cells (> 80%) (Yao et al. 2019). In contrast, more than 95% TCRαβ were observed in our study (Fig. 1e, f). This large difference may be due to differences in the origins of T cells. Whereas,

Table 1 Canonical cell markers in the cluster analysis

Cluster	Cell marker
Natural kill T cell	CD52, TXNIP, RPL34, RPS14, LGALS1, S100A6, MCM6, CDC25B
Progenitor cell	CDC20, NUF2, ASPM
Germ cell	EEF1A1, RPL28
T helper2 cell	IL13, IL5, TNFSF11, PTGDR2, IL4

the proportion of TCR-αβ decreased rapidly in activated DN-TILs.

Previous studies show that even in the context of tumor heterogeneity, TILs and DNTs can kill tumor cells effectively (Chambers and Neumann 2011) [41–43]. To investigate whether DN-TILs induced cytotoxicity against tumor cells, we cocultured DN-TILs with four cell lines from different tumors. Similar to the anti-tumor properties of DNTs and TILs, DN-TILs derived from gastric carcinoma showed cytotoxicity against cancer cells of different origins, including gastric carcinoma, pancreatic cancer, ovarian cancer and malignant melanoma (Fig. 2).

Notably, the cytotoxicity of DN-TILs derived from gastric cancer was not limited to gastric cancer (HGC-27), even shown in pancreatic cancer (PANC-1), melanoma (A375) and ovarian cancer (SKOV-3) (Fig. 2). It was assumed that DN-TILs should target a broad range of cancers in a donor-independent manner similar to DNT (Yao et al. 2019). The innate characteristic of recognizing antigenic targets independent of MHC causes TCR-$\gamma\delta^+$ DNT cells to recognize and kill different tumor strains (Gomes et al. 2010).

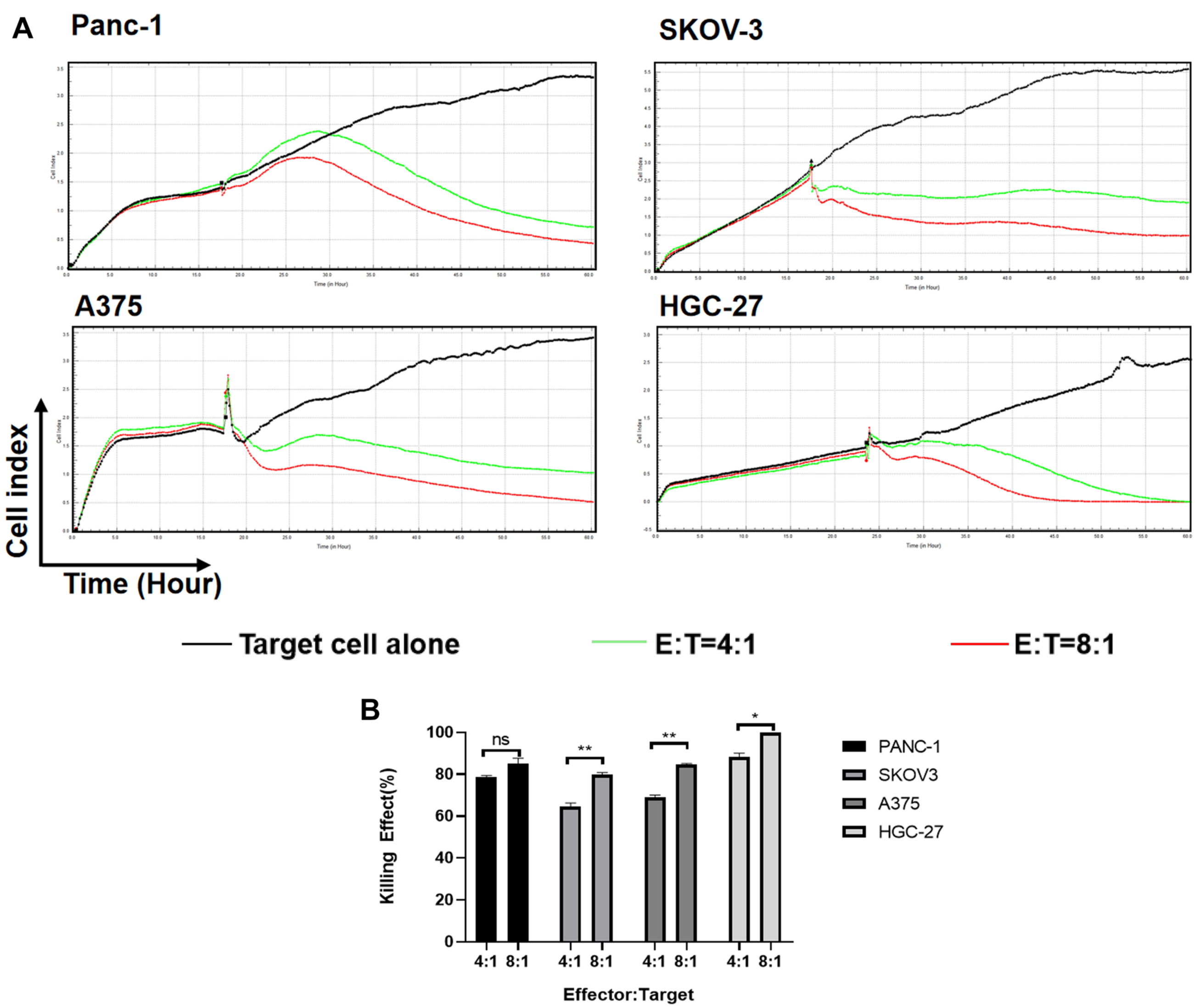


Fig. 2 Effect of DN-TILs on proliferation of Panc-1, SKOV-3, A375, and HGC-27. **A** suppression effect of DN-TILs on proliferation, cells were measured every 10 min terms by Real-Time Cell Analyzer (RTCA); **B** statistical chart of RTCA

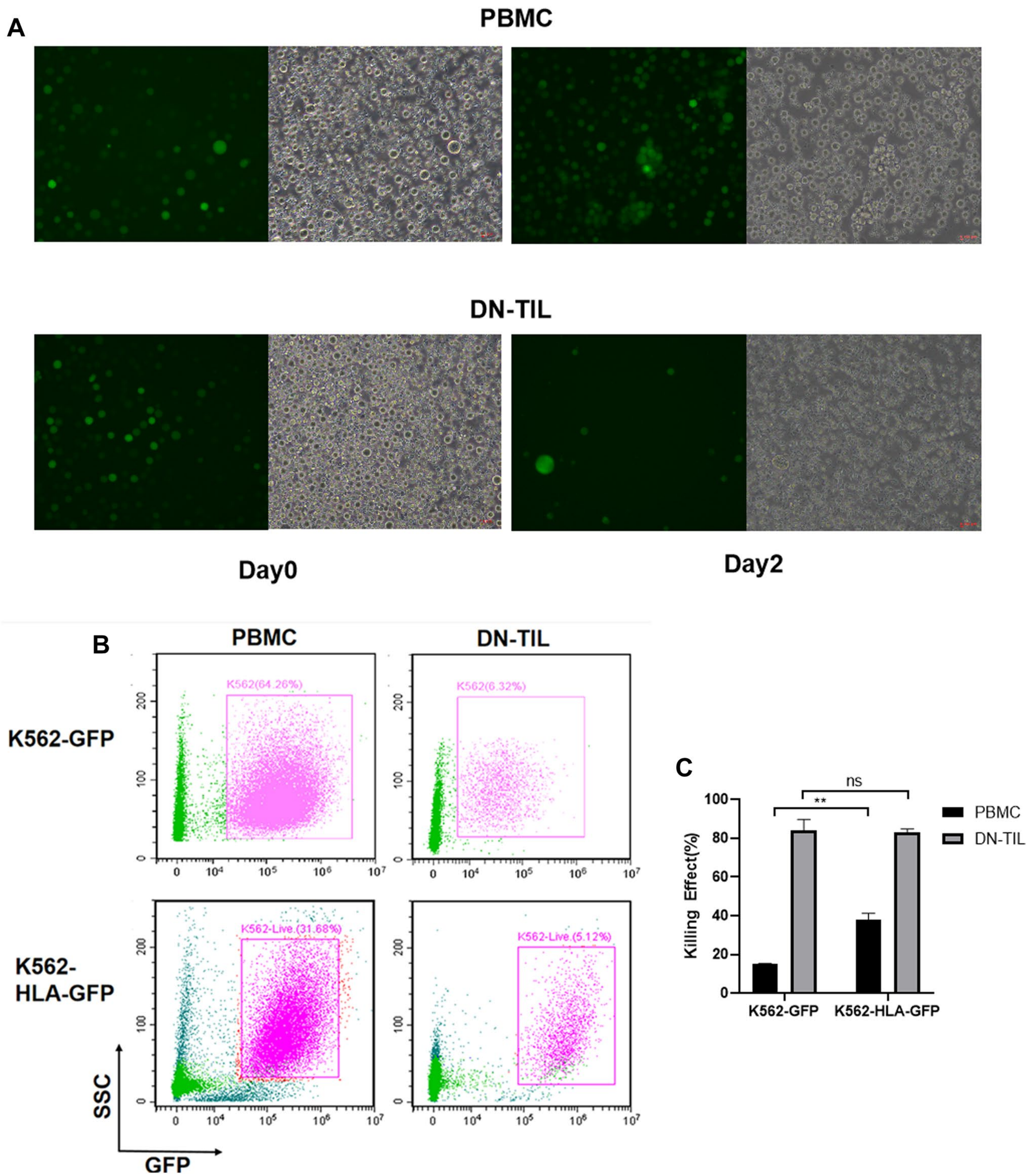


Fig. 3 Anti-proliferation effect of DN-TILs in MHC-independent way. **A** HLA-EGFP (green) levels in K562 cocultured with PBMC or DN-TILs (×400 magnification); **B** Living cells counted by flow cytometry in K562-HLA-EGFP and K562-EGFP; **C** statistical chart of differences. $^{**}P < 0.01$

Therefore, it was hypothesized that DN-TIL has the same nature in recognizing targets that were verified by transferring HLA to K562 in present study (Fig. 3). In addition, DNT cells expanded among total TILs from pancreatic and glioma patient tumors showed intracellular IFNγ and TNFα expression upon stimulation by autologous tumor [40, 41].

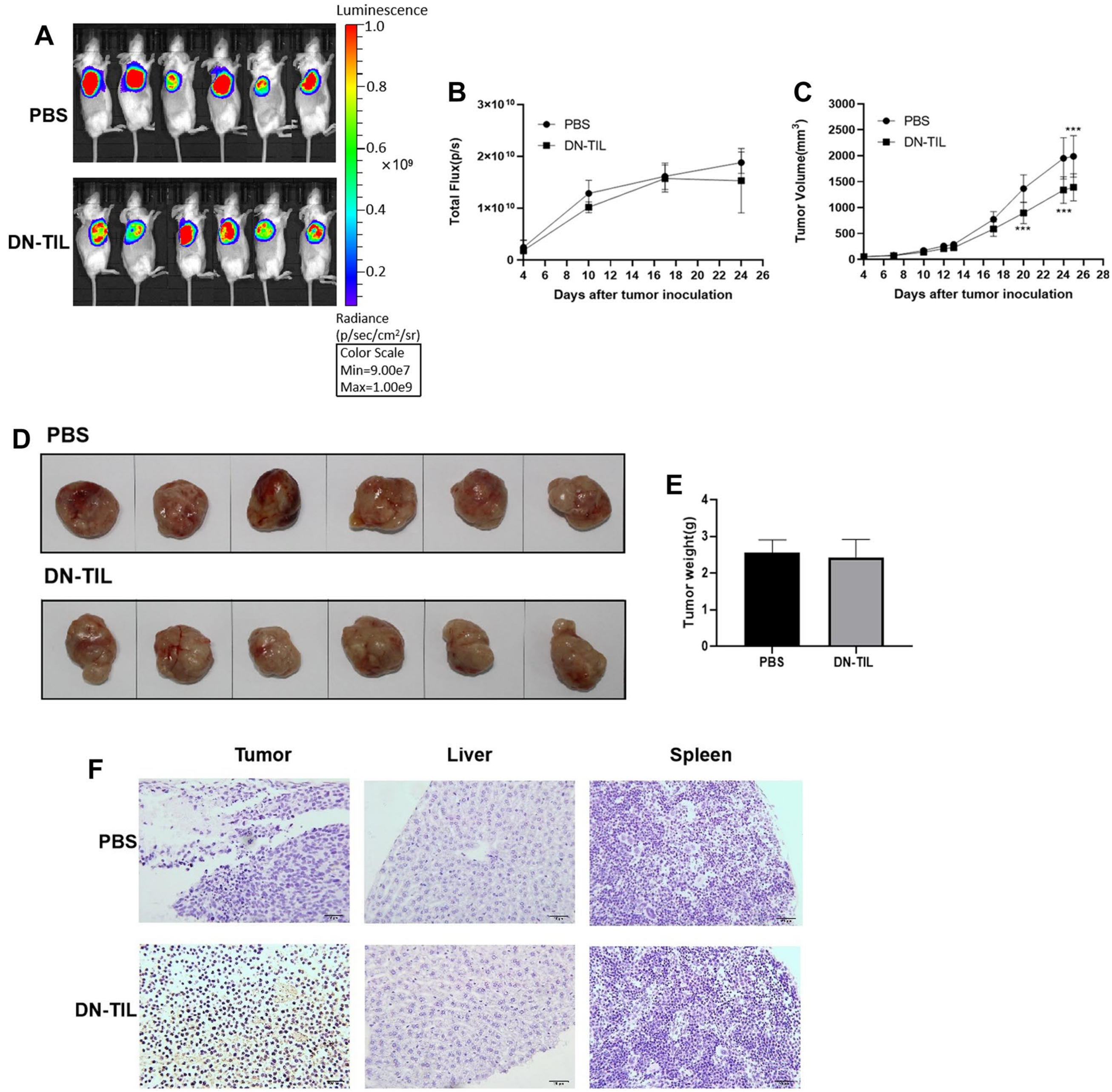


Fig. 4 Antitumor effect of DN-TILs in vivo. **A** The fluorescent signals of HGC-27 cells were monitored with an IVIS after DN-TILs injection; **B** Statistical chart of IVIS imaging; **C** tumor size monitoring after DN-TILs injection; **D** tumor images of xenograft tumor from different group; (E) Tumor weight measurement at 3 weeks after injection; **F** IHC staining of CD3 in xenograft tumors, liver and spleen (×400 magnification)

Collectively, these results show that DN-TILs may be poised with anti-tumor ability and share a similar cytotoxic pathway as DNTs and TILs.

In recent years, DNTs derived from healthy donor have been introduced into acute myeloid leukemia (AML) therapy basing on the donor-unrestricted manner (Chen et al. 2018). However, the application of DNT in solid tumors was limited due to the complex microenvironment and delivery difficulties. Therefore, the resent scientific challenge in the field is the use of CAR-DNT therapy to treat solid tumors (Lee et al. 2018). In contrast, our results showed that DN-TILs had superior tumor-homing ability and low off-target toxicity (Fig. 4). Similar characteristics endow TIL therapy unique advantages in treating solid tumors (Kazemi et al. 2022).

Nevertheless, broad application of TIL therapy has been limited due to the personalized approach, generating cellular products for individual patients "on-demand" using a patient's own cells (Chambers and Neumann 2011; Brenner 2012). Meanwhile, DNTs expanded from healthy volunteers under good manufacturing practice (GMP) conditions can be cryopreserved with long shelf life and reserved function in vitro and in vivo (Yao et al. 2019). Taken together, DN-TIL should be a promising therapeutic option for solid tumors.

There was an undeniable limitation in this study. DN-TILs were derived from only one patient who donated the resected tumor tissue to our hospital's specimen bank. Therefore, we cannot obtain more clinical information from the patient. Meanwhile, deriving DN-TIL requires fresh tissue, only one chance to obtain the tissue before sending the resected tumor to the cryopreserved bank. All of our experiments were based on limited cells that cannot support other hypothesis verification. In the future, we will focus on these specific cells to further explore the clinical significance.

Conclusion

We demonstrated for the first time that DN-TILs can be derived from gastric carcinoma. Our data show that DN-TILs are cytotoxic to different cancer cells in vitro and in vivo. DN-TIL cytotoxity was MHC-independent, similar to DNT, and had the same homing ability as TILs. These results highlight the effect of DN-TILs and the potential of DN-TIL cell therapy for the treatment of solid cancer.

Author contributions All authors contributed to the study conception or design. Clinical samples were provided by XG and QZ. Material preparation was performed by RX. Cell assays were performed by JL, CH and RH. Data collection and analysis were performed by JL, CH, YL and QX. JL, CH and QX contributed to the study conception and design. The first draft of the manuscript was written by JL and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript. JL and CH are equal first authors of this publication.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest Chen Huang is an employee of Shanghai Juncell Biotechnology Co., LTD. The other authors declare that they have no conflict of interest.

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